How much recorded intraoperative hypotension is arterial-line artefact, and does it bias the association with acute kidney injury? A waveform-referenced cohort study

Running title: Arterial-line artefact and intraoperative hypotension

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Key points

- In 3,249 patients with continuous arterial monitoring, 2.3% of monitored time was artefact (3.7% including the margins around artefact). With per-minute medians and a range filter, 0.7% of recorded minutes with mean arterial pressure (MAP) below 65 mmHg were spurious.
- In this cohort, ≥ 10 min of MAP below 65 mmHg was not independently associated with acute kidney injury (AKI). Artefact handling neither created nor removed an association, and artefact was not recorded preferentially in patients who developed AKI.
- Running-median smoothing and unfiltered area-under-threshold exposures distorted the measured hypotension. Over a third of the unfiltered area below 55 mmHg was artefact, half of it in 1% of patients.
- A recorded fall to 20–39 mmHg turned out to be artefact in 79% of cases when it had lasted 10 s, 57% at 30 s and 8% at 60 s. A sudden fall should prompt a look at the waveform, not a delay in treatment.
- Arterial artefact rose sixfold during the non-invasive cuff cycle, mostly as a non-pulsatile trace. This is consistent with occlusion of the cannulated arm.

Abstract

Background. Studies linking intraoperative hypotension to acute kidney injury (AKI) use monitor numbers that include arterial-line artefacts. It is unknown how much recorded hypotension is artefact, and whether artefact handling biases the association.

Methods. We analysed 3,249 adults in the open VitalDB dataset who had a 500 Hz arterial waveform. Each 2-s monitor value was labelled from the waveform, giving a waveform-referenced series. We compared this series with 420 retrospective processing pipelines (7 filters × 5 resolutions × 4 metrics × 3 thresholds). The primary estimand was the ratio of adjusted odds ratios (ROR) for KDIGO AKI with ≥ 10 min of mean arterial pressure (MAP) < 65 mmHg. It compared per-minute medians with a range filter against the reference, with an equivalence margin of 0.80–1.25.

Results. Of monitored time, 96.3% was artefact-free. With per-minute medians, 0.7% of recorded hypotensive minutes were spurious, against 3.3% with one instantaneous value per minute. Of 2,563 patients in the AKI analysis, 91 (3.6%) developed AKI. The adjusted odds ratio was 0.94 (95% CI 0.60–1.48) with the reference and 0.90 (0.57–1.42) with the pipeline. The ROR was 0.96 (0.86–1.06), which was equivalent. Exposure differed in 33 patients, none of whom developed AKI. Spurious hypotension was not associated with AKI. Arterial artefact rose from 4% to 26% during cuff measurement.

Conclusions. With per-minute medians and a range filter, artefact did not bias the association between hypotension and AKI in this cohort, in which the association was null. Running-median smoothing and unfiltered area-based exposures should be avoided.

Introduction

Intraoperative hypotension is associated with acute kidney injury (AKI), myocardial injury and death.[1–4] Thresholds derived from these studies, typically a mean arterial pressure (MAP) below 65 mmHg for more than a few minutes, now appear in consensus statements and quality metrics.[5] Almost all of this evidence comes from retrospective analyses of numbers stored by anaesthesia information management systems. These are one value per minute or less, taken from the monitor rather than from the waveform the clinician sees.

Invasive arterial pressure is the reference standard of these studies. Yet the stored numbers inherit every event that disturbs an arterial line:

- flushing;
- blood sampling through a stopcock;
- zeroing;
- a kinked or damped catheter;
- cuff inflation on the same limb.[6–9]

Earlier work established three things:

- choosing among published artefact filters changes the measured incidence of hypotension up to twofold, but changed the association with myocardial injury little;[10]
- coarser sampling inflates the measured burden of hypotension against a waveform reference, although its effect on outcome associations has only been extrapolated;[11]
- how hypotension is modelled changes the estimated odds ratios.[12]

Recent studies of the open VitalDB dataset relating arterial pressure to kidney injury applied a single cleaning rule or did not describe one.[13–15]

Four questions remain open: 1. When artefacts are identified from the waveform rather than from rules applied to the numbers, how much of recorded hypotension is artefact? 2. Do the artefact rule and the data resolution change the estimated association between hypotension and AKI when that association is estimated directly? 3. Is artefact misclassification *differential*? Arterial blood is sampled more often in sicker patients. If sampling artefact were recorded as hypotension preferentially in patients who later develop AKI, the error would bias published associations away from the null rather than towards it. 4. Which recorded low-pressure episodes are artefact, and how quickly can this be recognised?

We used the two parallel recordings in VitalDB, the 500 Hz waveform and the monitor's 2-s numbers, to build a waveform-referenced MAP series. We compared it with 420 pipelines that a retrospective study could plausibly use, in a pre-specified multiverse analysis.[16,17]

Methods

Design, data and reporting

We performed a retrospective cohort study of the open VitalDB dataset:[18]

- 6,388 non-cardiac surgical cases;
- Seoul National University Hospital, 2016–2017;
- anaesthesia records, laboratory results, monitor numerics (GE Solar 8000) and high-resolution waveforms.

The institutional review board approved data collection. The data are de-identified and public, so no further approval was needed. Reporting follows STROBE and RECORD.[19,20]

The analysis protocol was written before any exposure–outcome association was computed and was revised after an adversarial methods review. It was frozen by a tagged commit after the whole analysis had run end to end on a randomly permuted outcome. The Supplementary Protocol lists every amendment before the freeze, and every change and post hoc analysis after unblinding, with dates and reasons. Post hoc analyses are labelled as such throughout.

Participants

We included adults under general anaesthesia whose record contained the 500 Hz arterial pressure waveform (radial in 95%) and at least 30 min of reference-valid monitoring. Only the first eligible case per patient was used.

For the AKI analysis we further excluded:

- transplantation;
- patients without a baseline creatinine;
- baseline estimated glomerular filtration rate (eGFR, CKD-EPI 2021[21]) below 15 mL min^{-1} 1.73 m^{-2};
- patients without a creatinine measurement within 7 days after anaesthesia.

Waveform classification and the reference series

The monitor stores systolic, diastolic and mean arterial pressure every 2 s. This is the stream that anaesthesia information management systems store and that retrospective studies analyse. We used the waveform only to decide *whether* each 2-s monitor value was trustworthy, never to replace it, so calibration differences between monitor and waveform could not enter the comparison.

Each 2-s waveform window was labelled with the first category that applied:

- *zero*: mean below 15 mmHg and 2nd–98th centile range below 10 mmHg (transducer open to air or being zeroed);
- *high*: maximum above 250 mmHg or mean above 200 mmHg (fast flush);
- *non-pulsatile*: range below 8 mmHg, interdecile range below 3 mmHg, or no systolic upstroke (a rise above 8 mmHg within 100 ms) in the centred 6-s window. This covers a stopcock turned off to the patient and an occluded or kinked line;
- *noise*: minimum below –10 mmHg, or median absolute second difference ≥ 4 mmHg;
- *valid*: none of the above.

Because the monitor averages over several beats, windows from 2 s before to 8 s after any artefact run were labelled *guard*. Eight seconds was the 95th centile of the time the monitor took to rejoin the waveform after an artefact.

Thresholds were set on outcome-free data. A pulsatility criterion based on beat-to-beat autocorrelation was discarded before the freeze because it labelled atrial fibrillation as artefact. Figure 1 shows examples.

Alignment and monitoring period.

- *Lag*. The monitor numerics lag the waveform. The lag was estimated for each case by maximising the correlation between monitor MAP and the waveform window mean in valid windows. It was 4 s in 62% of cases and 6 s in 35%.
- *Monitoring period*. It ran from the first to the last uninterrupted 60 s of valid waveform within anaesthesia. It began a median of 9.9 min (IQR 6.3–15.8) after the start of anaesthesia, so most post-induction hypotension falls outside it.[22]

Reference series. The reference was the lag-aligned monitor MAP restricted to valid windows.

- Gaps of up to 3 min, bounded on both sides by at least 30 s of valid data, were bridged by linear interpolation between the medians before and after the gap. Bridging avoids a systematic loss of true hypotension that coincides with blood sampling.
- A minute was valid if at least half of its windows were valid or bridged.

Retrospective pipelines

A pipeline turns the monitor numbers into an exposure, using the numbers alone.

Filters reproduced published rules:

- F0: none;
- F1: MAP below 20 or above 250 mmHg removed;[13]
- F2: MAP below 30 or above 200 mmHg removed;[23]
- F3: systolic above 250, diastolic below 15, MAP below 35 or above 150, or pulse pressure below 15 mmHg;[11]
- F4: physiological consistency (MAP below 20 or above 200; systolic below 20 or above 300; systolic ≤ diastolic; pulse pressure below 5; diastolic below 5);
- F5: F4, then removal of values that differ from both neighbours by more than 30 mmHg;[24]
- F6: F4, then a 7-point running median.[10]

Resolutions:

- R1: every 2-s value;
- R2: one instantaneous value per minute;
- R3: the per-minute median;
- R4: a value charted every 5 min and held;
- R5: a value charted every 15 min and held.

R4 and R5 were evaluated at several phase offsets, and the estimates were averaged.

Exposures, each at thresholds of 55, 60 and 65 mmHg:

- cumulative minutes below the threshold, per 10 min, capped at 120 min;
- area under the threshold, per 100 mmHg·min, capped at the reference's 99th centile;
- time-weighted average below the threshold, in mmHg;
- the binary exposure "≥ 10 min below the threshold".

The reference was processed at every resolution in the same way. Each pipeline was compared with the reference at the same resolution, so the contrast isolates waveform-based cleaning. In total there were 420 pipeline specifications.

Outcomes and covariates

The primary outcome was AKI by the KDIGO creatinine criteria:[25] an increase of ≥ 0.3 mg dL^{-1} within 48 h, or ≥ 1.5 times baseline within 7 days. Baseline was the last creatinine within 30 days before surgery, otherwise the preoperative value.

Models were adjusted for:

- age and eGFR, as restricted cubic splines;
- sex, body mass index, ASA physical status ≥ 3 and emergency surgery;
- hypertension and diabetes;
- preoperative haemoglobin and albumin;
- department and surgical approach;
- log anaesthesia duration.

Vasopressors, fluids and blood loss were not included, because they may lie on the causal path. With 91 events and about 20 parameters there were about 4.5 events per parameter. We used Firth-penalised logistic regression,[26] and the two models that form each ratio share the same adjustment set.

Statistical analysis

Measurement error. Each recorded minute with MAP below 65 mmHg was classed against the reference at the same resolution:

- *spurious*: reference ≥ 65 mmHg;
- *confirmed*: reference < 65 mmHg;
- *indeterminate*: no valid reference. Indeterminate minutes were never counted as artefact.

Masked minutes were reference minutes below 65 mmHg that the pipeline recorded as ≥ 65 mmHg. For the binary exposure we report κ and the number of patients whose category differed. Proportions have patient-cluster bootstrap CIs.

Primary estimand. The primary estimand was the ratio of odds ratios, $ROR = OR_{pipeline} / OR_{reference}$, for AKI with the binary exposure "≥ 10 min below 65 mmHg". It compared F1 × R3 with the reference at R3, using the same adjustment set and the same patients. The ROR is unaffected by confounding that the two series share.

We pre-specified an equivalence margin of 0.80–1.25, the conventional margin symmetric on the log scale:

- an interval inside the margin: equivalent;
- an interval excluding 1, with a point estimate outside the margin: material change;
- anything else: inconclusive.

Because the two exposures differ in only a few patients, the percentile bootstrap interval is too narrow. Across 100 outcome permutations, in which the true ROR is 1, it covered 1 in only 51%. A rule written before the real outcome was examined therefore set every ROR-type interval to:

log ROR ± 1.96 × max(paired-bootstrap SE, SD over 200 outcome permutations).

We checked this interval by its coverage under independent permutations, and by simulating outcomes from the adjusted reference model with differential error placed at the margin (Supplementary Methods).

Secondary analyses.

- The same ROR for all 35 filter × resolution cells, with simultaneous 95% bands.
- The slope per 10 min below 65 mmHg.
- The full specification curve over metrics and thresholds.
- The effect of resolution on the reference itself.

Differential misclassification. We regressed AKI on log(1 + spurious minutes per monitored hour), adjusting for:

- the reference exposure (linear and spline terms);
- log monitored hours;
- the adjustment set.

We used the raw 2-s stream (F0 × R1), because at F1 × R3 spurious minutes were non-zero in too few patients (amendment A14, made before the freeze). The negative control was spurious hypertensive minutes: pipeline MAP above 120 mmHg while the reference was ≤ 100 mmHg.

Episodes. Each run of monitor MAP below 65 mmHg was classed as:

- *waveform-flagged*: ≥ 50% of its windows labelled artefact;
- *guard-dominated*;
- *physiological*.

We tabulated the probability that an episode was flagged by final duration and nadir. Post hoc, we also tabulated it by elapsed duration and nadir so far, which is what is known while an episode unfolds.

Validation of the reference.

- *Independent signals:*

  - cuff MAP within 60 s of an episode;
  - ECG and plethysmographic pulse rate during non-pulsatile windows and deep episodes;
  - the non-pulsatile fraction in atrial fibrillation.

- *Injection–recovery* of synthetic artefacts after Li et al.,[27] with amplitude and duration varied post hoc. The injections resemble the classifier's own rules, so this is an implementation check rather than independent validation.
- *A blinded AI rating* of 152 snippets (Supplementary Methods).

Sensitivity analyses. The pre-specified sensitivity analyses varied:

- the reference: guard length, pulse-pressure threshold, minute validity, bridging, and the waveform mean in place of the monitor value;
- the monitoring period;
- the adjustment set;
- the outcome: AKI stage ≥ 2; missing postoperative creatinine counted as no AKI (S4); transplantation included (S6).

S4 and S6 had been omitted by error and were run after unblinding. Post hoc analyses are listed with them in Table S5.

Analyses used Python 3.10 (NumPy, SciPy, pandas). The classifier, Firth regression, splines, bootstrap and permutation code were written by the author and unit-tested.

Results

Participants

Of 3,645 cases with an arterial waveform track, 3,434 had a retrievable waveform and 3,429 had a monitoring period. The descriptive cohort comprised 3,249 patients. Together they had 11,466 h of monitoring, a median of 193 min per patient (IQR 134–272).

The AKI analysis set comprised 2,563 patients, of whom 91 (3.6%) developed AKI (Figure S1; Table 1). Inclusion in the AKI set depended on postoperative creatinine testing. Of non-transplant patients in the descriptive cohort, the AKI set included:

- 96% of thoracic patients;
- 97% of urology patients;
- 85% of general surgery patients;
- 28% of gynaecology patients.

How much of the arterial record is artefact

Within the monitoring period, 96.3% of 2-s windows were valid. Artefact proper accounted for 2.3%:

- *high* 1.2%;
- *non-pulsatile* 0.7%;
- *noise* 0.2%;
- *zero* 0.07%.

Guard margins accounted for a further 1.4%. Inside the monitoring period the waveform was never missing: the analogue channel records continuously, and the period is bounded by valid data.

Artefact events were common:

Artefact type	Patients with at least one event	Events per monitored hour
Flush	98%	2.6
Noise	97%	2.7
Non-pulsatile run	90%	2.0
Zeroing	33%	0.2

In valid windows, monitor MAP and the waveform mean differed by more than 10 mmHg in a median of 0.5% of windows per patient.

Artefact around cuff measurements (Figure 2). In 2,579 patients with both arterial and cuff pressure, 14.0% of arterial windows in the 60 s before a new cuff value were labelled artefact, against 1.8% at other times (19,560 readings). The excess peaked 22 s before the value appeared, during the oscillometric cycle. At the peak 25.8% of windows were artefact, against 4.3% 4–5 min earlier.

The peak was almost entirely *non-pulsatile* trace: 22.4% against 1.7% at baseline. Flush labels barely rose (2.3% against 1.7%).

In post hoc analyses the excess was:

- **Present at regularly cycled readings**, which are probably automatic (12.2% of windows), as well as irregular ones (17.0%).
- **Larger with radial than with pedal or femoral catheters**, which an arm cuff cannot occlude: 13.9% against 1.8% at other times, compared with 9.7% against 3.4% ($p < 0.001$).
- **Larger with right than with left radial catheters.** Artefact preceded at least 10% of cuff readings in 60% of patients with right radial catheters and 31% of those with left.

The cuff arm is not recorded in VitalDB.

Spurious and masked hypotension

At the per-minute median resolution (R3) with filter F1, 0.66% (95% CI 0.58–0.74) of recorded minutes with MAP below 65 mmHg were spurious (Figure 3; Table S1). Spurious fractions were higher:

- with one instantaneous value per minute (R2): 3.3% (3.0–3.6), which is 4–10 times the R3 value depending on the filter;
- with the raw 2-s stream (R1): 2.0% (1.9–2.2).

Filters F3–F5 removed most spurious minutes at R1 and R3 (0.3–0.9%), but not at R2 (1.8–3.3%). The 7-point running median (F6) behaved differently from every other filter:

- it created spurious minutes: 7.5% at R3 and 9.9% at R2;
- it masked 20% (R3) to 24% (R2) of true hypotensive minutes.

Binary exposure. Against the reference at the same resolution, "≥ 10 min below 65 mmHg" was classified with κ 0.95–0.99 by every filter except F6 (Table S2). With F1 × R3, 38 of 3,249 patients (1.2%) changed exposure category, 32 of them upwards; 30 of the 38 had 5–10 reference minutes.

Resolution changed exposure prevalence more than any filter. In the reference series itself, 42.4% of patients were exposed at R3, 51.2% at R4 and 53.2% at R5.

The 7-point median applied to charted data erased most exposure: κ was 0.39 at R4 and 0.20 at R5.

Which recorded low-pressure episodes are artefact

Of 106,725 recorded episodes of MAP below 65 mmHg in 3,123 patients:

- 8,471 (7.9%) were waveform-flagged;
- 3,909 (3.7%) were guard-dominated;
- 94,345 (88.4%) were physiological.

Flagged episodes accounted for 4.3% of hypotensive minutes. The probability that an episode was flagged depended steeply on its final duration and nadir (Table 2):

- episodes lasting ≤ 60 s with a nadir below 40 mmHg were flagged in 65–90%;
- brief episodes (≤ 30 s) with a nadir of 40–54 mmHg were flagged in 59–65%;
- episodes with a nadir of 55–64 mmHg were flagged in ≤ 2.4%;
- episodes lasting more than 1 min with a nadir ≥ 40 mmHg were flagged in ≤ 2%.

Final duration is unknown while an episode is unfolding. For episodes still below 65 mmHg after a given time, the proportion flagged depended on the lowest pressure reached so far (post hoc, Table S3):

Nadir so far	10 s	30 s	60 s	120 s
20–39 mmHg	79%	57%	8%	2%
Below 20 mmHg	75%	59%	44%	29%

A simple rule, a fall of more than 20 mmHg from the preceding 30-s maximum, identified flagged episodes with sensitivity 0.89 and specificity 0.92 in a validation half. Its specificity was carried by trivial dips: among episodes with a nadir below 40 mmHg, specificity fell to 0.07.

Validation of the reference

Cuff MAP. Among deep episodes (nadir below 40 mmHg), a cuff MAP within 60 s was ≥ 70 mmHg in 81% (95% CI 79–82) of 2,081 flagged episodes, against 28% (24–32) of 443 physiological ones.

Pulse during flat or deep traces. During flagged deep episodes the plethysmographic pulse rate was in the physiological range in 98% of windows. During 153,662 non-pulsatile windows the ECG heart rate was 40–150 min^{-1} in 97.8%. Circulation therefore continued while the arterial trace was flat.

Atrial fibrillation. The non-pulsatile fraction was not higher in the 13 patients with preoperative atrial fibrillation (0.17% against 0.68%).

Injection–recovery (Table S8). The classifier detected every injected flush lasting at least 2 s or reaching 250 mmHg, every open-transducer level up to 25 mmHg, and every sampling episode. It rarely or never detected:

- short flushes of 200 mmHg or less;
- over-damping;
- partial occlusion, except severe;
- slow drift;
- ringing.

Blinded AI rating (secondary; Table S9). Agreement with the classifier was 97.7% (κ 0.95).

Association with acute kidney injury

The reference association. For ≥ 10 min below 65 mmHg the odds ratio for AKI was:

- crude: 1.65 (1.09–2.51);
- age, eGFR, ASA and emergency adjusted: 1.46 (0.95–2.25);
- fully adjusted: 0.94 (0.60–1.48).

Removing anaesthesia duration from the adjustment set did not change the fully adjusted estimate. For 55 mmHg the fully adjusted odds ratio was 1.95 (0.89–4.27), and per 10 min below 65 mmHg it was 1.08 (0.99–1.17) (Table S11).

Primary estimand. With F1 × R3 the fully adjusted odds ratio was 0.90 (0.57–1.42). The ROR was 0.96 (95% CI 0.86–1.06; permutation p = 0.42): equivalent.

The contrast rested on the 33 patients whose exposure category differed: 28 moved up and 5 moved down. None of them developed AKI, against 1.6 expected from the adjusted reference model (probability of none 0.19).

Had AKI occurred in k of the 28 upward-moved patients, the ROR would have been:

k	0	1	2	3	4	5
ROR	0.96	1.01	1.07	1.13	1.20	1.27

Equivalence would have been lost only from about five cases onwards (Table S7).

In simulations from the adjusted reference model, the interval rule declared equivalence:

- in 82% of datasets without differential error;
- in 5% of datasets with differential error that placed the median ROR at 1.24.

The interval covered 1 in 97–100% of independent outcome permutations.

All 35 filter × resolution cells (Table 3; Figure 4). No 95% interval or simultaneous band excluded 1. Twenty-five cells were equivalent pointwise and 19 under simultaneous bands.

Four cells at R2 sat within 0.002 of the upper margin, two on each side of it. The inconclusive cells were:

- the raw 2-s stream (R1);
- instantaneous per-minute sampling (R2);
- the 7-point median.

With any filter other than F6, the point estimates ranged from 0.94 to 1.12. Coarser resolution of the reference itself did not detectably change the odds ratio: relative to R3, the ratios were 0.93–1.10, with intervals between 0.76 and 1.42.

Where the reference association was stronger (post hoc). At 55 mmHg the binary ROR was imprecise:

- F1: 1.28 (0.81–2.02);
- F3: 1.10 (0.84–1.45).

At 60 mmHg it was 0.95–0.98, with intervals of 0.79–1.15. The slope per 10 min below 65 mmHg was 1.09 with F1 × R3 against 1.08 with the reference. The difference in log-odds was 0.007 (−0.005 to 0.019).

Exposure metric (Figures S2–S4; Table S4). Across all 420 specifications the median ROR was 0.99 (IQR 0.95–1.01).

- Minute-based exposures were insensitive to processing (RORs 0.92–1.02).
- Area-based exposures below 55 mmHg were not. Without a filter, 37% of the total area below 55 mmHg was artefact, half of it in 1% of patients. With F1 the share was 13%, and with F3 it was none. The corresponding RORs were large but imprecise (post hoc; time-weighted average, F0: 0.30, 0.06–1.53).
- For 90 specifications, mostly area-based, the pipeline and the reference differed in whether their odds ratio was statistically significant.

Sensitivity analyses (Table S5). The primary ROR ranged from 0.94 to 0.97 across the reference variants, adjustment variants, S4, S6, the exclusion of gynaecology and the disease-risk score. All intervals lay within 0.83–1.10. The adjustments for blood loss and for creatinine count, and AKI stage ≥ 2, have point estimates only. The exceptions were the waveform-mean reference, 0.74 (0.48–1.16), whose monitor–waveform calibration differences the primary design deliberately avoided, and the numerics-defined period, 0.91 (0.75–1.11).

Differential misclassification (Table S6). Spurious hypotension was unrelated to how often arterial blood was sampled (Spearman ρ 0.005, p = 0.81). Adjusted for reference exposure, the odds ratio for AKI per SD of spurious minutes on the raw stream was 0.93 (0.74–1.16); per unit of log(1 + minutes per hour) it was 0.59 (0.12–2.86). The negative control, fitted separately, gave 1.16 (0.95–1.42) per SD. These estimates are imprecise and cannot exclude modest differential error, but they gave no sign of it.

Discussion

Principal findings

In 3,249 patients with a high-resolution arterial waveform, artefacts were frequent events but a small part of recorded hypotension. With the processing most retrospective studies use, a per-minute median and a range filter, fewer than 1% of hypotensive minutes were spurious. The exposure category changed in 1.2% of patients.

Two features of the cohort define what our main result can and cannot say:

- **The canonical exposure was not associated with AKI after adjustment.** For ≥ 10 min below 65 mmHg the adjusted odds ratio was 0.94, although the crude odds ratio was 1.65.
- **Non-differential error had almost nothing to act on.** When the reference association is null, non-differential misclassification cannot attenuate it. The equivalence of the pipeline and the reference (ROR 0.96) therefore tests mainly *differential* error, the scenario in which artefact recorded preferentially in patients destined for AKI would manufacture an association.

That scenario was not present:

- the 33 patients whose category changed included no AKI case;

- spurious hypotension was unrelated to blood sampling and to AKI;
- the equivalence verdict held under every sensitivity analysis, including the two pre-specified analyses we initially omitted.

Where an association was present (55 mmHg, and the slope per 10 min), the pipeline estimates were close to the reference but too imprecise to establish equivalence.

Three processing choices mattered more than the filter did:

- **The 7-point running median** created spurious hypotension and masked a fifth of true hypotension. On charted data it erased most exposure.
- **Unfiltered area-based exposures below 55 mmHg** were dominated by zero and flush values in a few patients: over a third of the area was artefact.
- **Resolution** classified a fifth more patients as exposed, without a detectable change in the odds ratio.

Comparison with previous work

Filters and resolution. Pasma et al. compared eight filters on per-minute data. The filter changed the measured incidence of hypotension up to twofold but changed the association with myocardial injury little.[10] Jacobsson et al. showed, against a waveform reference, that coarse sampling inflates measured hypotension, and extrapolated an effect on outcomes.[11] Our study adds three things:

- an external, waveform-referenced comparator at every resolution;
- a direct estimate of the outcome effect, with a pre-specified equivalence criterion;
- a test for differential error.

We agree with both studies on duration-based exposures. Unlike Pasma et al., we found that the 7-point median distorts exposure substantially, even where the ratio of odds ratios happened to stay near 1. The AWAKE trial showed that continuous arterial monitoring changes the *measured* and *treated* hypotension at induction.[28] Our resolution results are the retrospective counterpart: the same patients look different at different sampling rates.

Cuff and arterial pressure. Wax et al.,[29] and more recently a preprint from the MOVER cohort,[30] treated disagreement between cuff and arterial pressure as clinical information. We found a measurement mechanism that generates such disagreement: arterial artefact rose sixfold during the cuff cycle, mostly as a non-pulsatile trace. Several features point to occlusion of the cannulated arm in a subset of patients:

- the excess was larger with radial than with lower-limb catheters;
- it was larger on the right than on the left;
- it was present at regularly cycled readings;
- flush labels barely rose.

It was not the whole story. A smaller excess with pedal and femoral catheters suggests other activity around cuff cycles, such as handling of the arm or the line. Studies of discordance should exclude arterial values from the cuff cycle, or show that their results survive that exclusion.

Artefact detection. Earlier work compared algorithmic and machine-learning detection with human annotation of stored values,[7,31] and a scoping review found no agreed reference for arterial artefact.[32] Our classifier uses the waveform only to decide whether a monitor value can be trusted. It can be reproduced on any dataset that stores both.

Implications

For investigators:

- Use a per-minute median with a range filter, or a rule such as F3. Report the resolution as carefully as the threshold.
- Do not use running-median smoothing to estimate exposure.
- Do not compute area- or time-weighted exposures below 55 mmHg from unfiltered data.
- In paired multiverse comparisons, check the operating characteristics of the intervals before unblinding, at the null and at the decision margins. Permutation alone checks only the null. Without this check, our percentile intervals would have covered the truth in half of null datasets.

For clinicians and alarm design:

- Most very brief, very deep recorded falls proved to be artefact in retrospect. Within the first 10–30 s, a fall to 20–39 mmHg was artefact more often than not. By 60 s, more than nine in ten were real.
- A fall below 20 mmHg remained artefact in a third of cases even after 2 min.
- These are retrospective probabilities. They do not justify delaying treatment. A sudden fall should prompt an immediate look at the arterial waveform, the plethysmograph and end-tidal carbon dioxide.
- A system could use them to *annotate* a probable artefact, never to *suppress* an alarm. The simple drop rule we derived cannot even separate deep artefacts from deep real events (specificity 0.07 below 40 mmHg).
- Where possible, place the cuff on the limb opposite an arterial line, consistent with standard guidance on arterial pressure measurement.[9]

Strengths and limitations

Strengths:

- The reference comes from the waveform recorded alongside the numbers, and it never replaces monitor values.
- The protocol was frozen after a permuted dry run. Every later change is documented.
- The validity of the reference was checked against independent signals: cuff pressure, pulse oximetry and ECG.
- The data are public and the code is released.

Limitations:

- **No human rater adjudicated the reference.** The AI rater shares its model family with the classifier's designer and sees only the waveform. A 400-snippet set with a hidden key is released for human adjudication, and it should show the monitor value as well as the waveform.
- **The classifier cannot see some important errors.** These are:
  - levelling errors and zero drift (about 7 mmHg per 10 cm, which matters in lateral and tilted positions);
  - most over-damping;
  - partial occlusion;
  - very short flushes.

  Our reference is therefore *waveform-screened*, not error-free.
- **The monitoring period started a median of 10 min after the start of anaesthesia.** It therefore missed most post-induction hypotension,[22] which limits comparison with studies that use cuff values from induction onwards.
- **Single centre.** The data come from one centre in 2016–2017, mainly general and thoracic surgery, with a low incidence of creatinine-defined AKI (3.6%).
- **AKI ascertainment.** AKI was ascertained from creatinine only, without urine output, and depended on postoperative testing, which varied by department. Haemodilution may have hidden some AKI.
- **Few events.** With 91 events, the precision of each odds ratio is modest. The ratio of two estimates in the same patients is far more precise, but our equivalence result rests on few discordant patients and may not transfer to cohorts with more artefact or a stronger association.
- **Charting was simulated.** Simulated charting is a random sample of monitor values, not the selective charting of real records.[6]
- **Practice-dependent frequencies.** Artefact frequencies depend on local practice, so the absolute numbers will not transfer. The mechanisms and the direction of the resolution and smooth-

ing effects should.

Conclusions

Arterial-line artefacts were frequent events but a small part of recorded hypotension. In this cohort, ≥ 10 min of MAP below 65 mmHg was not independently associated with AKI. With per-minute medians and a range filter, artefact handling neither created nor removed an association, and artefact was not recorded preferentially in patients who developed AKI.

Investigators should:

- avoid running-median smoothing;
- avoid unfiltered area-based exposures;
- report resolution as carefully as thresholds.

A brief, very deep recorded fall is often artefact. It should prompt an immediate look at the waveform, not a delay in treatment. Arterial artefact clusters in the cuff cycle, consistent with occlusion of the cannulated arm.

Declarations

Ethics. This is a secondary analysis of the de-identified, publicly released VitalDB dataset. Collection of the data was approved by the institutional review board of Seoul National University Hospital, as described in the data descriptor.[18] No further approval or consent was required.

Funding. None.

Competing interests. None declared.

Data and code availability. VitalDB is openly available at https://vitaldb.net. The analysis code, the frozen protocol with all amendments, the permuted dry run, aggregate results and the 400-snippet adjudication set are available at [REPOSITORY URL]. The adjudication key is available from the author on request, to preserve blinding of future raters.

Use of artificial intelligence. The author used an AI system (Claude, Anthropic) for substantial parts of this work, under the author's direction:

- writing the analysis code;
- an adversarial review of the methods before the protocol was frozen;
- three simulated peer reviews of the draft (statistical, clinical, signal processing), which led to the post-unblinding changes listed in the Supplementary Protocol;
- drafting the manuscript.

An instance of the same AI system was also used as the blinded secondary rater of waveform snippets, as described in the Methods. The author checked the code, outputs and text, and takes full responsibility for the content. The AI system is not an author.

Author contribution. O.T. conceived and designed the study, performed the analysis, interpreted the results and wrote the manuscript.

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Figure legends

Figure 1. Recorded hypotension that is waveform-flagged artefact. Each panel shows the 500 Hz arterial waveform (grey) and the lag-aligned 2-s monitor MAP (orange) around the start of a recorded MAP below 65 mmHg (time 0). Shaded windows are labelled artefact or guard. From top to bottom:

- a transducer open to air (zeroing);
- a stopcock turned off to the patient during blood sampling;
- a fast flush followed by recovery;
- a non-pulsatile trace with a cuff reading within 3 min.

Figure 2. Arterial artefact around non-invasive cuff measurements, by artefact type. The plot shows the percentage of 2-s arterial windows with each label, by time relative to the moment a new cuff value appears (14,307 readings with a complete −300 to +30 s window).

Figure 3. Spurious minutes among recorded minutes with MAP below 65 mmHg, by filter and resolution: R1, 2-s stream; R2, one value per minute; R3, 1-min median. Orange: spurious (reference ≥ 65 mmHg), with patient-cluster bootstrap 95% CI. Blue: indeterminate (no valid reference), never counted as artefact.

Figure 4. Ratio of odds ratios for AKI (≥ 10 min below 65 mmHg) for every filter × resolution combination, each against the reference at the same resolution.

- Points: ratio of odds ratios.
- Thin lines: 95% CI, log ROR ± 1.96 × max(bootstrap SE, permutation-null SD).
- Wide bands: simultaneous 95% bands over 35 cells.
- Grey: equivalence margin 0.80–1.25.

Figure S1. Flow of patients.

Figure S2. Specification curve: ratios of odds ratios for all 420 pipeline specifications, sorted. The lower panel marks the filter, resolution and threshold.

Figure S3. Probability that a recorded episode is waveform-flagged, by final duration and nadir (the data of Table 2).

Figure S4. Ratio of odds ratios at the 1-min median resolution by filter, exposure metric and threshold (post hoc display). Orange: ROR < 1; blue: ROR > 1.

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Table 1. Patient characteristics

Characteristic	Descriptive cohort	AKI set: no AKI	AKI set: AKI
n	3249	2472	91
Age, years	61 (51–69)	62 (54–70)	63 (56–72)
Female	1444 (44.4)	1053 (42.6)	19 (20.9)
BMI, kg m^{-2}	23.0 (20.8–25.2)	23.1 (20.9–25.2)	22.6 (20.4–26.0)
ASA physical status ≥ 3	492 (15.1)	283 (11.4)	28 (30.8)
Emergency surgery	413 (12.7)	266 (10.8)	19 (20.9)
Hypertension	1085 (33.4)	835 (33.8)	42 (46.2)
Diabetes	398 (12.2)	318 (12.9)	18 (19.8)
Baseline eGFR, mL min^{-1} 1.73 m^{-2}	95 (82–104)	95 (83–103)	94 (74–104)
Preoperative haemoglobin, g dL^{-1}	13.0 (11.6–14.2)	13.1 (11.8–14.3)	11.7 (10.4–13.6)

Characteristic	Descriptive cohort	AKI set: no AKI	AKI set: AKI
General surgery	2017 (62.1)	1430 (57.8)	67 (73.6)
Thoracic surgery	1001 (30.8)	943 (38.1)	17 (18.7)
Gynaecology / urology	231 (7.1)	99 (4.0)	7 (7.7)
Open approach	1318 (40.6)	908 (36.7)	61 (67.0)
Anaesthesia duration, h	3.8 (2.8–5.2)	3.8 (2.8–4.9)	4.3 (3.2–6.3)
Monitored time, min	193 (134–272)	188 (135–258)	223 (157–346)
Reference minutes MAP < 65 mmHg	7 (1–19)	7 (1–18)	11 (4–39)
Reference ≥ 10 min MAP < 65 mmHg	1377 (42.4)	1021 (41.3)	49 (53.8)

Values are median (IQR) or n (%). Reference = waveform-verified monitor MAP, 1-min median.

Table 2. Probability that a recorded low-MAP episode is waveform-flagged artefact

Nadir duration	<=10 s	11-30 s	31-60 s	1-2 min	2-5 min	>5 min
<20 mmHg	79 (77–81) [1569]	77 (74–80) [905]	65 (58–71) [201]	43 (34–52) [122]	23 (16–31) [150]	11 (4–21) [116]
20-39 mmHg	67 (64–70) [755]	90 (89–92) [1813]	75 (70–80) [386]	12 (6–18) [125]	1 (0–3) [232]	4 (1–6) [248]
40-54 mmHg	59 (56–63) [1391]	65 (61–69) [1508]	13 (10–16) [661]	2 (1–4) [903]	1 (0–1) [1578]	1 (0–2) [1177]
55-64 mmHg	2 (2–3) [66981]	1 (1–2) [13525]	1 (0–2) [5286]	1 (0–2) [3346]	0 (0–1) [2867]	1 (0–1) [880]

Episode: contiguous run of monitor MAP < 65 mmHg. Cells: % flagged (patient-cluster bootstrap 95% CI) [number of episodes].

Table 3. Every filter × resolution: measurement error and its effect on the AKI association

F	R	Spurious %	Masked %	Exposed % P / R	κ	ROR (95% CI)	Simult. band
F0	R1	2.3	1.0	44.1 / 42.3	0.96	1.12 (0.92–1.36)	0.82–1.54
F0	R2	3.4	1.8	46.5 / 44.5	0.95	1.06 (0.89–1.25)	0.80–1.39
F0	R3	0.8	1.3	43.4 / 42.4	0.97	0.99 (0.88–1.11)	0.82–1.20
F0	R4	–	–	53.1 / 51.2	0.95	0.94 (0.86–1.04)	0.81–1.10
F0	R5	–	–	54.4 / 53.2	0.96	0.99 (0.92–1.07)	0.87–1.12
F1	R1	2.0	0.8	43.7 / 42.3	0.96	1.09 (0.91–1.31)	0.81–1.47
F1	R2	3.3	1.8	46.3 / 44.5	0.96	1.06 (0.89–1.25)	0.80–1.39
F1	R3	0.7	1.1	43.2 / 42.4	0.98	0.96 (0.86–1.06)	0.80–1.14
F1	R4	–	–	52.9 / 51.2	0.95	0.95 (0.87–1.03)	0.82–1.09
F1	R5	–	–	54.2 / 53.2	0.97	1.00 (0.93–1.08)	0.88–1.13
F2	R1	1.8	0.6	43.5 / 42.3	0.97	1.12 (0.94–1.34)	0.83–1.50
F2	R2	3.3	1.7	46.3 / 44.5	0.96	1.06 (0.90–1.25)	0.81–1.39

F	R	Spurious %	Masked %	Exposed % P / R	κ	ROR (95% CI)	Simult. band
F2	R3	0.5	0.9	43.0 / 42.4	0.98	0.97 (0.87–1.08)	0.82–1.15
F2	R4	–	–	53.0 / 51.2	0.95	0.94 (0.87–1.02)	0.82–1.08
F2	R5	–	–	54.3 / 53.2	0.97	1.01 (0.93–1.08)	0.89–1.13
F3	R1	0.6	0.4	42.5 / 42.3	0.98	1.05 (0.93–1.19)	0.86–1.28
F3	R2	3.1	1.6	46.1 / 44.5	0.96	1.07 (0.91–1.26)	0.81–1.40
F3	R3	0.3	0.8	42.9 / 42.4	0.98	0.98 (0.89–1.08)	0.83–1.15
F3	R4	–	–	53.1 / 51.2	0.95	0.95 (0.88–1.03)	0.83–1.08
F3	R5	–	–	54.3 / 53.2	0.97	1.01 (0.94–1.09)	0.90–1.14
F4	R1	0.9	0.6	42.8 / 42.3	0.98	1.03 (0.91–1.16)	0.84–1.26
F4	R2	3.3	1.7	46.4 / 44.5	0.96	1.06 (0.89–1.25)	0.80–1.39
F4	R3	0.4	0.9	42.9 / 42.4	0.98	0.97 (0.88–1.07)	0.82–1.14
F4	R4	–	–	53.3 / 51.2	0.95	0.94 (0.86–1.02)	0.82–1.08

F	R	Spurious %	Masked %	Exposed % P / R	κ	ROR (95% CI)	Simult. band
F4	R5	–	–	54.4 / 53.2	0.97	1.00 (0.93–1.08)	0.89–1.13
F5	R1	0.9	0.6	42.8 / 42.3	0.98	1.03 (0.91–1.16)	0.84–1.26
F5	R2	1.8	0.8	45.5 / 44.5	0.97	1.00 (0.83–1.20)	0.74–1.34
F5	R3	0.3	0.7	42.8 / 42.4	0.98	0.97 (0.88–1.07)	0.82–1.14
F5	R4	–	–	51.8 / 51.2	0.96	0.95 (0.87–1.03)	0.82–1.09
F5	R5	–	–	52.1 / 53.2	0.95	1.03 (0.95–1.12)	0.90–1.19
F6	R1	3.0	3.5	42.4 / 42.3	0.98	0.99 (0.89–1.09)	0.84–1.17
F6	R2	9.9	24.0	36.1 / 44.5	0.82	1.13 (0.83–1.54)	0.68–1.88
F6	R3	7.5	19.9	36.2 / 42.4	0.86	1.22 (0.92–1.61)	0.77–1.92
F6	R4	–	–	20.3 / 51.2	0.39	1.54 (0.97–2.44)	0.72–3.28
F6	R5	–	–	11.0 / 53.2	0.20	1.18 (0.67–2.07)	0.46–2.99

F, filter: F0 none; F1 MAP 20–250; F2 MAP 30–200; F3 Jacobsson rule; F4 physiological consistency; F5 F4 + neighbour rule; F6 F4 + 7-point running median. R, resolution: R1 2-s stream; R2 one in-

stantaneous value per minute; R3 1-min median; R4 5-min chart; R5 15-min chart. P / R, pipeline / reference. Spurious: pipeline minute < 65 mmHg with reference ≥ 65. Masked: share of reference minutes < 65 that the pipeline records ≥ 65; neither is defined for charted resolutions. Exposure: ≥ 10 min with MAP < 65 mmHg; exposure prevalence and κ from the descriptive cohort (charted resolutions at phase offset 0). ROR (ratio of odds ratios, pipeline / reference at matched resolution) from the AKI set; 95% CI = log ROR ± 1.96 × max(paired-bootstrap SE, permutation-null SD); simultaneous band: bootstrap max-|z| critical value over all 35 cells applied to the same SE.

Figures

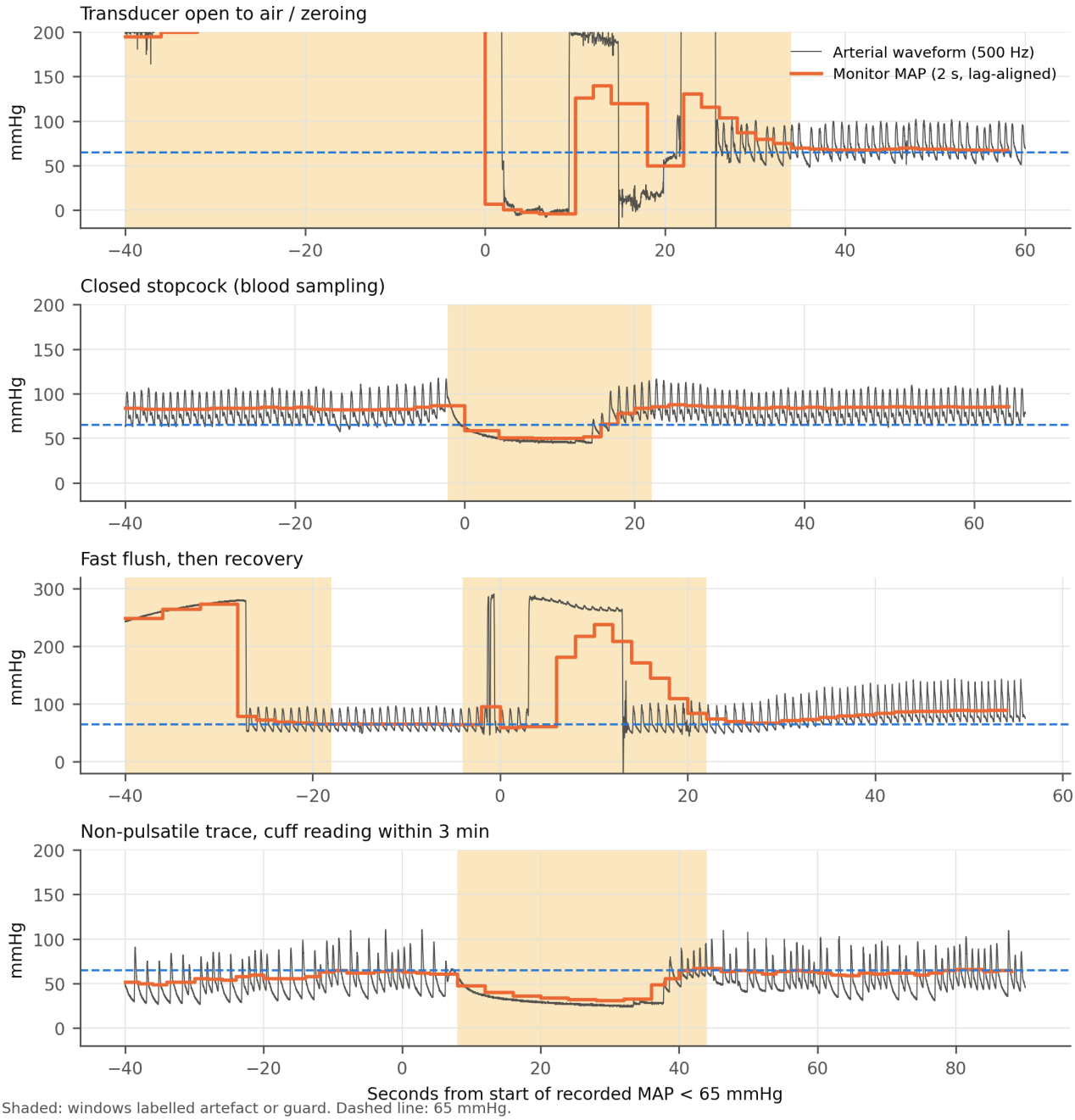


Figure 1: Recorded hypotension that is waveform-flagged artefact: waveform and monitor MAP.

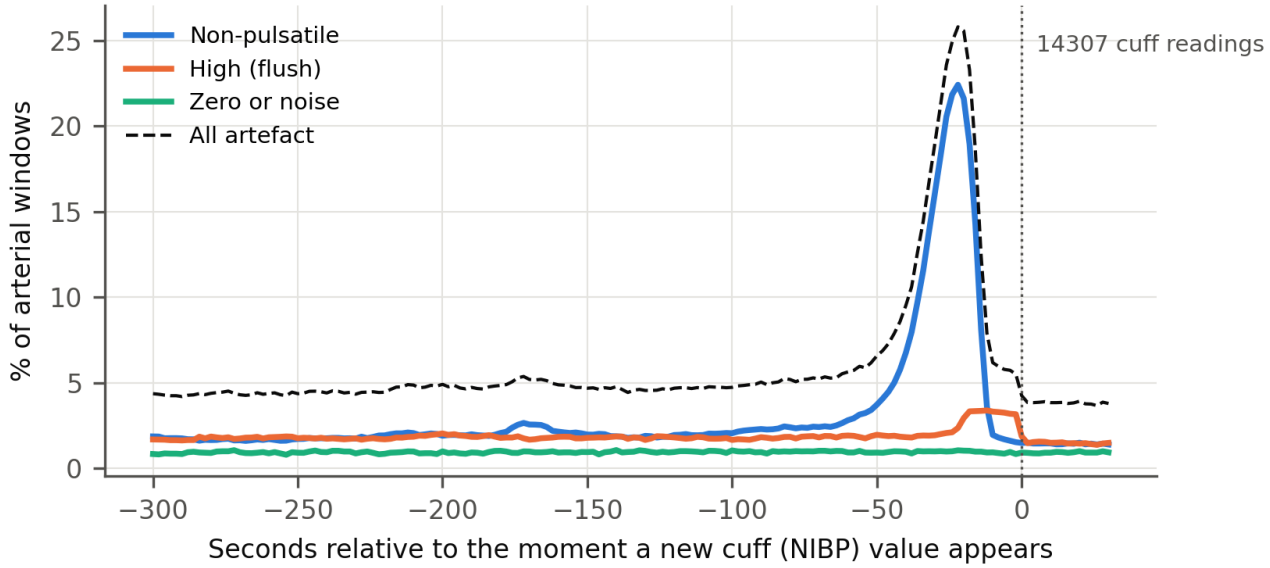


Figure 2: Arterial artefact around non-invasive cuff measurements, by artefact type.

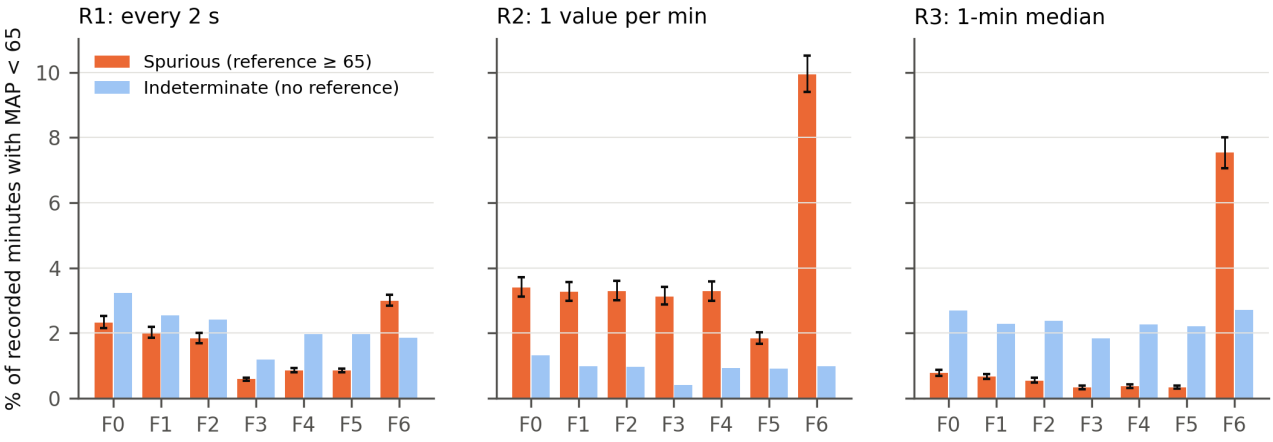
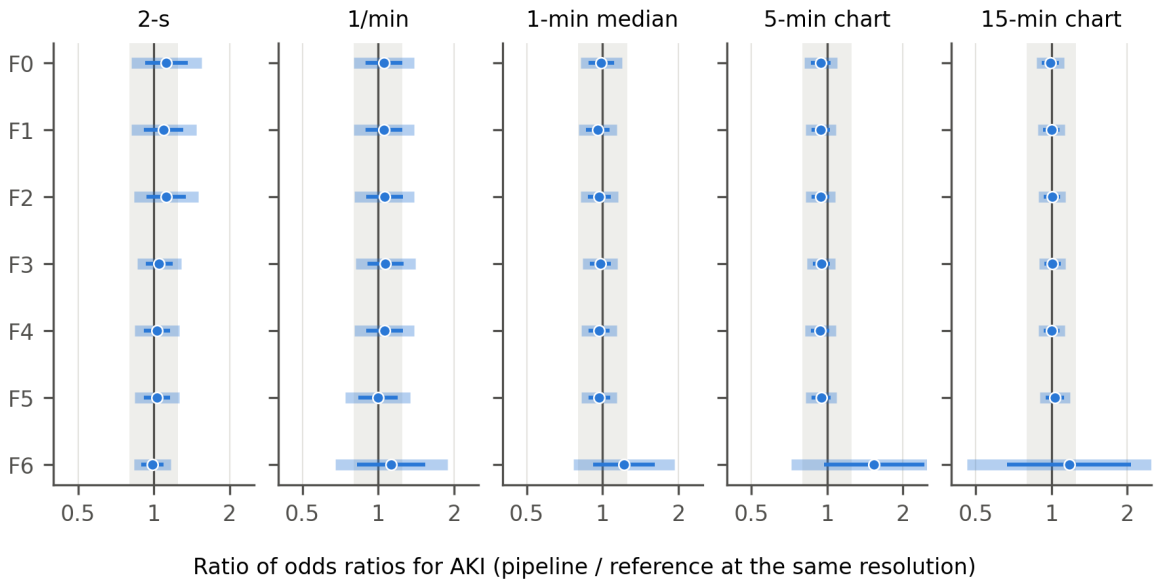


Figure 3: Spurious minutes among recorded minutes with MAP < 65 mmHg, by filter and resolution.



Thin line: 95% CI; wide band: simultaneous 95% band over 35 cells; grey: equivalence margin 0.80–1.25.

Figure 4: Ratio of odds ratios for AKI for every filter × resolution combination.